

Computer Vision Final Report

Street View House Numbers Recognition

A comparative study of YOLO26 and traditional HOG+SVM pipelines for robust digit detection in natural scenes.

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The Challenge: Digits in the Wild

Detecting house numbers in natural scenes is significantly more difficult than reading clean academic datasets (like MNIST).

- 📦 **Complex Backgrounds:** Clutter, architectural elements, and decorations.
- ⚙️ **Illumination Variations:** Shadows, extreme glare, and poor lighting.
- 📷 **Digit Overlap:** Characters physically touching or closely stacked.
- 📹 **Real-world Noise:** Blurry captures from moving vehicles (Street View).



Project Objectives



Model Implementation

Deploy a YOLO26-based object detection framework for single-stage, real-time inference on the SVHN dataset.



Data Pipeline

Build a conversion tool to transform the SVHN .mat annotation format into YOLO-compatible bounding box data.



Evaluation

Empirically evaluate multiple training configurations and establish a baseline comparison with HOG+SVM.



Post-Processing

Design a spatial grouping algorithm to reconstruct full, multi-digit house numbers from individual detections.

Methodology: Modern vs. Traditional

Baseline Method

HOG + SVM

A classical computer vision pipeline implemented for comparative evaluation.

Feature Extraction: Uses Histogram of Oriented Gradients to capture local edge orientations.

Classification: Employs a Support Vector Machine to classify extracted features.

Limitation: High false positive rate on background structures; struggles with low contrast.

Proposed Method

YOLO26 Architecture

A single-stage deep learning detector that performs localization and classification simultaneously.

Efficiency: Predicts bounding boxes and probabilities in a single forward pass.

Robustness: Learns hierarchical visual features directly from large-scale data.

Advantage: Exceptional balance between real-time inference speed and detection accuracy.

System Architecture & Data Flow



1. Data Prep

Convert 600k+ SVHN instances from .mat to YOLO format.
Normalize coordinates.



2. Training

GPU-accelerated YOLO26 training with mosaic augmentation and SGD optimization.



3. Inference

Individual digit detection using Non-Maximum Suppression (NMS) to filter overlaps.



4. Post-Process

Sort by X-coordinate and evaluate spatial proximity to reconstruct full numbers.

Inference Strategy: Reconstruction

The algorithm relies on the natural reading order, assuming digits of the same house number are horizontally aligned and spatially clustered.

$$\Delta X < Th$$

Horizontal Proximity

Step-by-Step Grouping Logic

Step 1 (Detect): YOLO model predicts individual bounding boxes [X, Y, W, H, C].

Step 2 (Sort): Order bounding boxes by their X-coordinate (from left to right).

Step 3 (Group): Evaluate horizontal distance (Delta X) and vertical alignment (Delta Y).

Step 4 (Concatenate): Merge valid grouped digits into a single string (e.g., [7] → [3] → [2] becomes "732").

Experimental Setup & Configurations

Hardware: Intel i5-14600K, RTX 3060 (12GB GPU), 32GB RAM. Framework: Ultralytics YOLO.

Configuration	Image Size	Batch Size	Epochs	mAP@0.5
Config A (Baseline)	640 × 640	32	10	0.890
Config B (Optimized)	320 × 320	128	30	0.904

Key Insight: Longer training (30 epochs) compensates for reduced resolution. Because SVHN digits are relatively large, the drop to 320px forced the network to learn robust generalized features rather than fine pixel details.

Training Metrics & Convergence

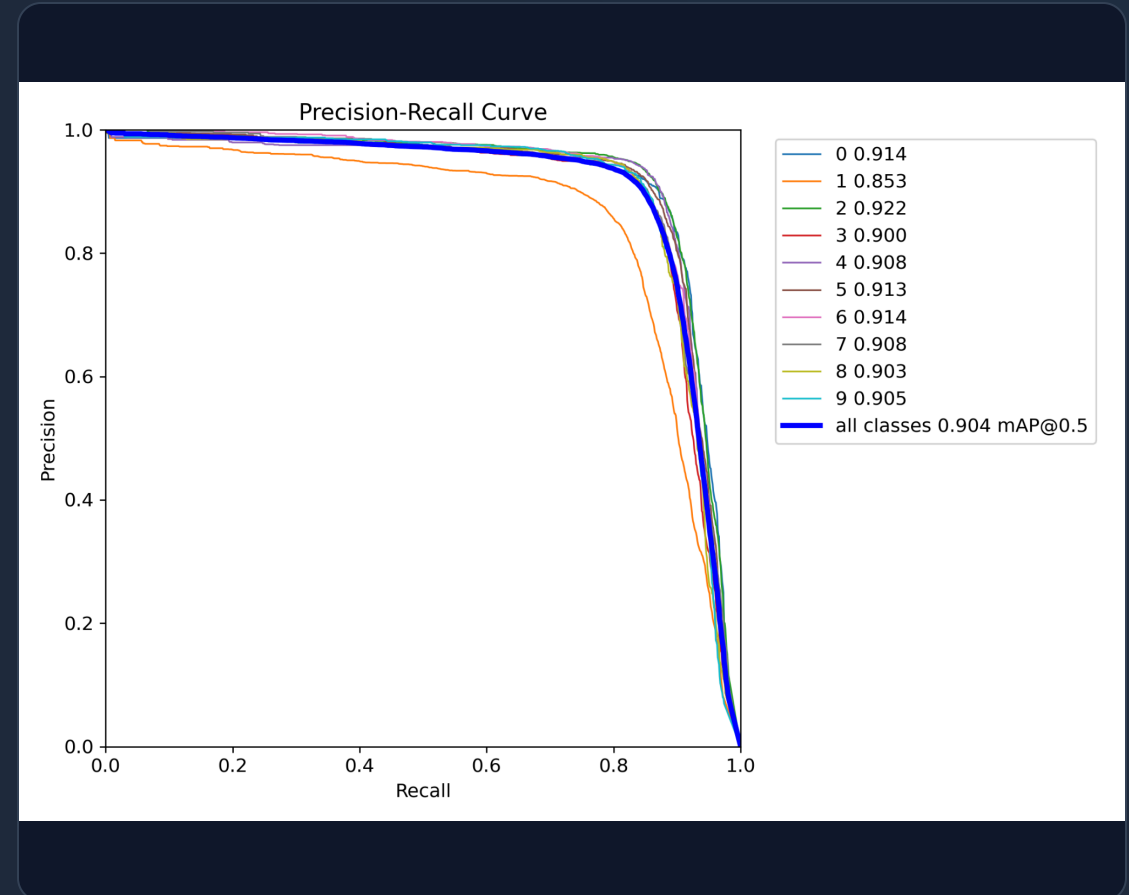
Precision-Recall Curve

The precision-recall relationship of the trained detector on the validation dataset (Figure 6).

Trade-off Balance: The curve demonstrates the optimal balance between precision and recall across various confidence thresholds.

High Performance: The area under the curve (mAP@0.5) reaches a competitive 0.904, indicating excellent overall detection capability.

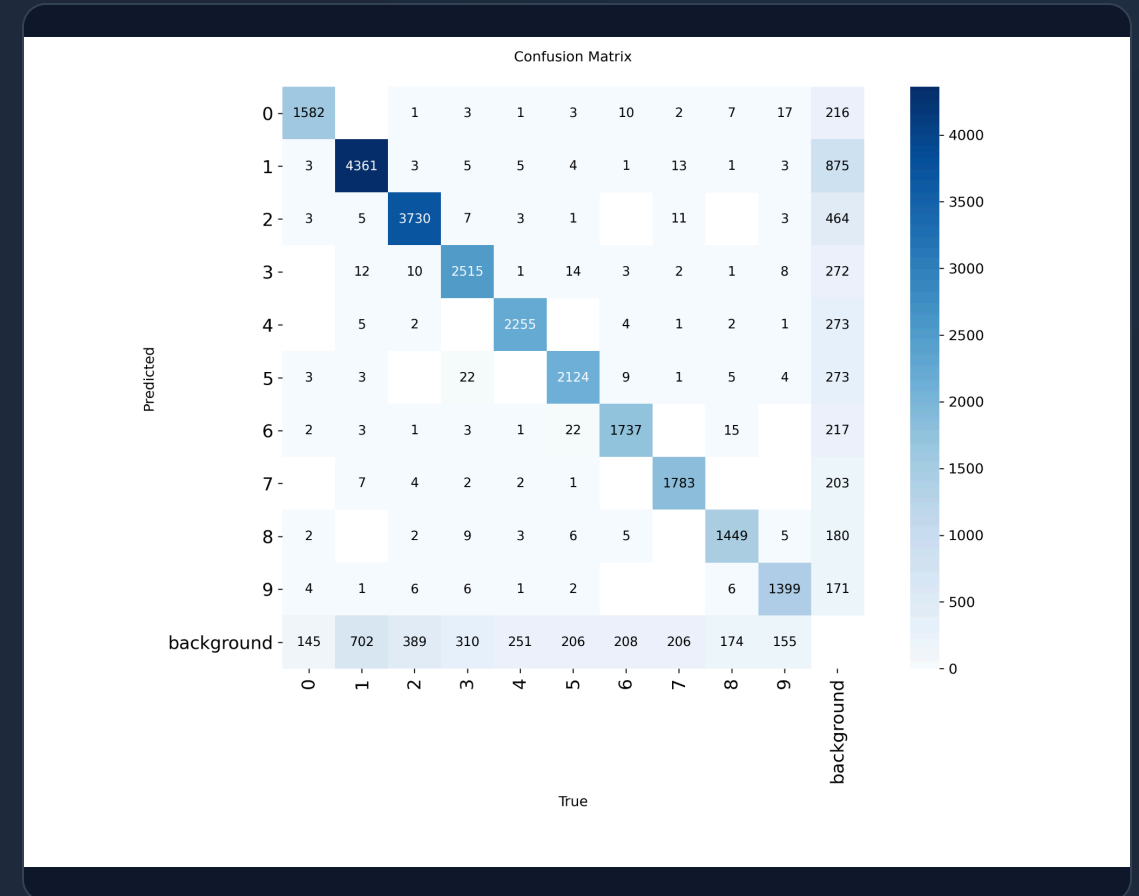
Stability: The precision remains high across most recall levels before dropping steeply, proving the model is highly confident in its true positive predictions.



Confusion Matrix Analysis

Error Analysis

- ✓ **Main Diagonal (True Positives):** Highly pronounced, indicating that most classes (especially 2, 5, and 6) are classified with extreme accuracy.
- ⚠ **Common Confusions:**
 - Digit '1' is frequently confused with the Background (False Positives).
 - Digit '6' is occasionally confused with '8' or '0' under blurry conditions.
- 💡 **Conclusion:** Background noise is the primary factor reducing Recall, especially for objects with straight vertical edges.



Qualitative Results (Inference Examples)

Example detection results produced by the YOLO-based house number detector (Figure 7).



Overall Detection Performance



YOLO26 significantly outperforms the traditional HOG+SVM baseline in both accuracy and robustness, maintaining high precision across varying confidence thresholds.

Parameter Sensitivity Analysis

To optimize the final inference pipeline, extensive parameter tuning was conducted to balance precision, recall, and false positives.

 **Confidence Threshold (Set to 0.3):** A threshold of 0.3 provided the best balance. Higher thresholds missed moderate-confidence digits, while lower thresholds introduced false detections from background structures.

IoU Threshold (Set to 0.2): Testing IoU from 0.2 to 0.6 showed minimal impact because SVHN digits rarely overlap significantly. 0.2 was chosen for strict NMS filtering.

 **Inference Resolution (Set to 960-1280px):** While training succeeded at 320px, inference at lower resolutions caused missed detections of small digits. High resolution ensures complete multi-digit reconstruction without heavy computational penalty.

Edge Cases & The '1' Anomaly

Missed Detections

Example cases where the model fails to detect one or more digits in a house number (Figure 10).

- ⚠️ **The '1' Bottleneck:** Digit '1' has the lowest recall (0.762). Its narrow structure is easily confused with vertical background edges or poles.
- 🚫 **Occlusion & Size:** The detector struggles when digits are partially hidden by physical objects or are extremely small.
- 🔦 **Low Contrast:** Failures occur when digits blend into the background (e.g., dark text on dark walls).



Real-Time Inference Speed

~53

Milliseconds per image

Breakdown (544x1280 resolution)

Preprocessing: 4.2 ms

Model Inference: 47.7 ms

Post-processing (Grouping): 1.5 ms

Validated on an RTX 3060 edge-equivalent setup, the pipeline operates comfortably in real-time, making it immediately viable for autonomous navigation deployments.

Edge Cases & Limitations



Missed Detections

The model occasionally drops digits that are:

- Partially occluded by physical objects.

- Extremely small relative to the image frame.

- Suffering from severe low background contrast.



Classification Confusion

Analysis of the confusion matrix reveals minor blending between visually similar digits:

- Distinguishing '4' from '6' in blurry conditions.

- Class imbalances in the SVHN dataset act as the primary hurdle to reaching 99%+ recall.

Conclusion & Future Roadmap



Project Summary

Successfully implemented an end-to-end YOLO26 pipeline. Achieved competitive 0.904 mAP with highly optimized spatial post-processing at ~53ms/image.



Refinement

Implement multi-scale inference during testing to solve small-digit edge cases and reduce missed detections caused by varying distances.



Deployment

Optimize further via TensorRT for deployment on low-power edge devices (e.g., NVIDIA Jetson Nano) for live smart city integration.

Questions?

Thank you for your attention.

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